



School of Computation, Information and Technology - Informatics  
Technical University of Munich

Bachelor's Thesis in Informatics

# **A Comparison of No-Swap-Regret Algorithms**

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**Ein Vergleich von No-Swap-Regret-Algorithmen**

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# Abstract

No-regret learning connects online decision making with equilibrium concepts in repeated games. When all players achieve vanishing external regret, empirical play approaches coarse correlated equilibrium (CCE), while vanishing internal or swap regret leads to the stronger notion of correlated equilibrium (CE). This thesis studies how these theoretical distinctions appear at finite horizons under different feedback models and reward environments.

A unified experimental framework is used to compare external-regret baselines, regret-matching procedures, and several swap-regret algorithms under full-information and bandit feedback. The experiments include two one-player reward environments and self-play in Rock–Paper–Scissors (RPS) and Rock–Paper–Scissors–Lizard–Spock (RPSLS). All learners are evaluated using the same external-, internal-, and swap-regret metrics. In self-play, the empirical joint-action distribution is additionally evaluated by its  $L_1$  distance to the CE and CCE sets.

The experiments show that the distinction between external and stronger regret notions is clearly visible at finite horizons in several settings. External-regret learners can achieve low external regret while retaining substantially larger internal and swap regret, whereas algorithms designed for stronger deviation classes control these measures more effectively. However, the no-swap-regret algorithms themselves show different finite-horizon behavior, and no single method dominates across all experiments. The separation between regret classes also depends on the tested reward sequence: it is particularly clear under the adaptive historical-frequency adversary and much weaker under the independent lazy random walk. In self-play, these regret differences are reflected in equilibrium convergence: external-regret learning can approach CCE while remaining farther from CE, whereas stronger regret control is generally accompanied by closer convergence toward CE.

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# 1 Introduction

## 1.1 Motivation

Online learning studies how a decision maker can adapt its behavior over time without knowing future rewards in advance. A common objective is to achieve low regret: after many rounds, the learner should not systematically perform worse than suitable alternatives chosen in hindsight.

Many standard online-learning algorithms focus on external regret, where the learner is compared with the best fixed action. This is a natural and useful benchmark, but it does not capture all possible improvements. In particular, a learner may perform well compared with every fixed action while still benefiting from systematically replacing one chosen action by another. Internal and swap regret address such action-dependent deviations and therefore provide stronger performance guarantees [1, 2].

These stronger guarantees are especially relevant in repeated games. Vanishing external regret leads to coarse correlated equilibrium (CCE) [3], whereas vanishing internal or swap regret leads to the smaller set of correlated equilibria (CE) [1, 2, 4]. However, stronger guarantees also require more sophisticated learning procedures. Reduction-based swap-regret algorithms, for example, may maintain several external-regret learners and repeatedly compute stationary distributions.

This raises a practical question: how much difference do these stronger guarantees make over finite horizons? Theoretical regret bounds describe asymptotic or worst-case behavior, but they do not directly show whether the additional complexity of no-swap-regret algorithms produces a clear empirical advantage in a given experiment. Different algorithms may also react differently to the available feedback and to the structure of the environment.

The aim of this thesis is therefore to compare several external-, internal-, and swap-regret algorithms experimentally. The comparison considers both full-information and bandit feedback, as well as one-player environments and repeated-game self-play. Besides regret itself, the self-play experiments also examine whether differences between the algorithms are reflected in convergence toward CE and CCE.

## 1.2 Research Questions

The thesis focuses on the following questions:

1. How do the considered regret-minimizing algorithms differ empirically at finite horizons?

2. How do these differences change across feedback models and environments?
3. How are these differences reflected in convergence toward CE and CCE in self-play?

### 1.3 Scope and Contributions

The experiments consider finite action spaces and rewards in  $[0, 1]$ . Both full-information and bandit feedback are studied. The algorithms are evaluated in one-player environments and in repeated-game self-play.

The thesis compares external-regret learners, regret-matching procedures, and several no-swap-regret methods using the same regret metrics. In the self-play experiments, empirical play is additionally evaluated by its distance to the CE and CCE sets.

The thesis does not introduce a new learning algorithm or regret bound. Its main contribution is a unified implementation and empirical comparison of existing methods, with particular attention to their finite-horizon behavior across different feedback models and environments.

### 1.4 Thesis Structure

Chapter 2 introduces the regret notions and their connection to CE and CCE. Chapter 3 describes the implemented algorithms, and Chapter 4 presents the experimental setup.

Chapter 5 reports the experimental results. Chapter 6 discusses the main observations and limitations, and Chapter 7 summarizes the findings.

## 2 Theoretical Background

### 2.1 Online Learning Setting

#### 2.1.1 Interaction Protocol

We consider a finite action set

$$[K] = \{1, \dots, K\}.$$

At round  $t$ , the learner chooses a probability distribution

$$p_t \in \Delta_K = \left\{ p \in \mathbb{R}_{\geq 0}^K : \sum_{i=1}^K p(i) = 1 \right\},$$

samples an action  $I_t \sim p_t$ , and receives reward  $r_t(I_t)$ , where

$$r_t = (r_t(1), \dots, r_t(K)) \in [0, 1]^K.$$

Randomization allows the learner to use mixed strategies rather than selecting a single action deterministically. In adversarial settings, this reduces the predictability of the learner's actions and can prevent an adversary from systematically exploiting a deterministic strategy. Under bandit feedback, randomization also provides a mechanism for exploring actions whose rewards are otherwise unobserved.

The learner chooses  $p_t$  before observing the feedback of round  $t$ . The reward vector  $r_t$  may depend on the previous interaction history, but not on the learner's fresh random draw  $I_t$  [5, 6].

Conditional on  $p_t$  and  $r_t$ , the expected reward in round  $t$  is

$$\mathbb{E}[r_t(I_t) \mid p_t, r_t] = \langle p_t, r_t \rangle.$$

This quantity will later be used to define strategy-weighted regret.

Throughout this thesis, rewards rather than losses are used. For rewards in  $[0, 1]$ , losses can equivalently be defined as

$$\ell_t(i) = 1 - r_t(i),$$

since

$$r_t(j) - r_t(i) = \ell_t(i) - \ell_t(j).$$

#### 2.1.2 Feedback Models

Two feedback models are considered.

**Full-information feedback.** After choosing an action, the learner observes the complete reward vector  $r_t$ . It can therefore use the rewards of all actions when updating its strategy.

**Bandit feedback.** The learner observes only the reward  $r_t(I_t)$  of the selected action, while the rewards of the remaining actions are hidden. Algorithms may therefore need to estimate information about unobserved actions from the available feedback.

## 2.2 Regret Notions

Regret compares the learner's cumulative reward with the reward that could have been obtained by applying alternative decisions to the same sequence of reward vectors. Different classes of alternatives give rise to external, internal, and swap regret [1, 2, 5].

For randomized learners, we distinguish between realized and strategy-weighted regret. Realized regret is computed from the actions actually sampled by the learner. Strategy-weighted regret instead averages each one-round deviation gain according to the mixed strategy  $p_t$ .

### 2.2.1 External Regret

External regret compares the learner with the best fixed action in hindsight:

$$R_T^{\text{ext}} = \max_{j \in [K]} \sum_{t=1}^T (r_t(j) - r_t(I_t)).$$

The corresponding strategy-weighted quantity is

$$\bar{R}_T^{\text{ext}} = \max_{j \in [K]} \sum_{t=1}^T (r_t(j) - \langle p_t, r_t \rangle).$$

A no-external-regret algorithm aims to achieve sublinear external regret, so that the average regret vanishes as the horizon grows:

$$\frac{R_T^{\text{ext}}}{T} \rightarrow 0.$$

### 2.2.2 Internal Regret

External regret compares against a single fixed action, regardless of which actions the learner actually selected. Internal regret instead considers action-dependent replacements.

For two actions  $i, j \in [K]$ , consider replacing every occurrence of action  $i$  by action  $j$ , while leaving all other actions unchanged. The realized replacement regret is

$$R_T^{i \rightarrow j} = \sum_{t=1}^T \mathbf{1}\{I_t = i\} (r_t(j) - r_t(i)).$$

Its strategy-weighted analogue is

$$\bar{R}_T^{i \rightarrow j} = \sum_{t=1}^T p_t(i) (r_t(j) - r_t(i)).$$

Internal regret is the largest gain obtainable from one such replacement:

$$R_T^{\text{int}} = \max_{i,j \in [K]} R_T^{i \rightarrow j},$$

and analogously,

$$\bar{R}_T^{\text{int}} = \max_{i,j \in [K]} \bar{R}_T^{i \rightarrow j}.$$

This action-dependent deviation is the regret notion underlying regret matching and its connection to correlated equilibrium [1, 3, 4].

### 2.2.3 Swap Regret

Swap regret generalizes internal regret by allowing a separate replacement for every source action. Let

$$\phi : [K] \rightarrow [K]$$

be a mapping from played actions to replacement actions. The realized swap regret is

$$R_T^{\text{swap}} = \max_{\phi: [K] \rightarrow [K]} \sum_{t=1}^T (r_t(\phi(I_t)) - r_t(I_t)).$$

Because the replacement  $\phi(i)$  can be chosen independently for each source action,

$$R_T^{\text{swap}} = \sum_{i=1}^K \max_{j \in [K]} R_T^{i \rightarrow j}.$$

Similarly, the strategy-weighted version is

$$\bar{R}_T^{\text{swap}} = \sum_{i=1}^K \max_{j \in [K]} \bar{R}_T^{i \rightarrow j}.$$

### 2.2.4 Relations Between the Regret Notions

External and internal deviations are special cases of swap deviations. Therefore,

$$R_T^{\text{ext}} \leq R_T^{\text{swap}}, \quad R_T^{\text{int}} \leq R_T^{\text{swap}}.$$

Furthermore,

$$R_T^{\text{swap}} \leq K R_T^{\text{int}},$$

because swap regret is the sum of at most  $K$  maximal source-action replacement regrets. Thus, for fixed  $K$ , no-internal-regret and no-swap-regret are asymptotically equivalent, although their dependence on  $K$  may differ [2].

The same relations hold for the corresponding strategy-weighted quantities.

The three deviation classes can be summarized as follows:

| Regret notion | Deviation                                    | Non-trivial deviations |
|---------------|----------------------------------------------|------------------------|
| External      | Replace all actions by one fixed action      | $K$                    |
| Internal      | Replace one action $i$ by one action $j$     | $K(K-1)$               |
| Swap          | Choose a replacement for every source action | $K^K - 1$              |

The increasingly rich deviation classes explain why low swap regret represents a stronger requirement than low external regret. They also lead to different equilibrium guarantees in repeated games.

## 2.3 Equilibrium Concepts

Consider a finite normal-form game with players  $i \in N$ , action sets  $A_i$ , and payoffs  $u_i(a) \in [0, 1]$ . After  $T$  rounds, the empirical distribution of joint actions is

$$\hat{\mu}_T(a) = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{a_t = a\}.$$

This distribution records the frequency with which each joint action profile occurs during repeated play.

### 2.3.1 Coarse Correlated Equilibrium

A distribution  $\mu$  is a coarse correlated equilibrium (CCE) if no player can improve by committing in advance to one fixed alternative action [3, 7]. For every player  $i$  and action  $a'_i \in A_i$ ,

$$\sum_a \mu(a) (u_i(a'_i, a_{-i}) - u_i(a)) \leq 0.$$

The deviation is fixed before observing the action assigned to the player. This is directly analogous to the fixed-action deviations considered by external regret.

### 2.3.2 Correlated Equilibrium

A distribution  $\mu$  is a correlated equilibrium (CE) if no player can improve by replacing one recommended action with another [8]. For every player  $i$  and every pair  $a_i, a'_i \in A_i$ ,

$$\sum_{a_{-i}} \mu(a_i, a_{-i}) (u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i})) \leq 0.$$

Unlike in a CCE deviation, the alternative action may depend on the recommended action. This corresponds to the action-dependent replacements used in internal regret.

Every correlated equilibrium is also a coarse correlated equilibrium:

$$\text{CE} \subseteq \text{CCE},$$

but the converse does not generally hold.

## 2.4 Regret and Equilibrium Convergence

The connection between regret minimization and equilibrium follows directly from the corresponding deviation inequalities [1, 3, 4].

For player  $i$  in a repeated game, the reward of action  $a_i$  at round  $t$  can be written as

$$r_{i,t}(a_i) = u_i(a_i, a_{-i,t}),$$

where  $a_{-i,t}$  denotes the actions selected by the other players.

Suppose every player has realized external regret at most  $\varepsilon_T T$ , where  $\varepsilon_T \rightarrow 0$ . Then, for every player  $i$  and every fixed deviation  $a'_i$ ,

$$\sum_a \hat{\mu}_T(a) (u_i(a'_i, a_{-i}) - u_i(a)) \leq \varepsilon_T.$$

Therefore,  $\hat{\mu}_T$  is an  $\varepsilon_T$ -CCE and approaches the CCE set as  $T$  grows.

For internal regret, suppose every realized replacement regret is at most  $\varepsilon_T T$ . Then, for every pair  $a_i, a'_i$ ,

$$\sum_{a_{-i}} \hat{\mu}_T(a_i, a_{-i}) (u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i})) \leq \varepsilon_T.$$

Thus, the empirical distribution is an  $\varepsilon_T$ -CE and approaches the CE set as  $T$  grows. Since

$$R_T^{\text{int}} \leq R_T^{\text{swap}},$$

vanishing realized swap regret implies the same conclusion [2].

The relationship relevant to the self-play experiments can therefore be summarized as

$$\text{vanishing external regret} \implies \text{CCE},$$

whereas

$$\text{vanishing internal or swap regret} \implies \text{CE}.$$

The experiments later evaluate this connection both through the players' regret and through the distance of the empirical joint-action distribution to the CE and CCE sets. The exact distance measure is introduced in Chapter 4.

## 3 Regret-Minimizing Algorithms

This chapter presents the regret-minimizing algorithms used in the experiments. All algorithms are existing methods from the literature; no new learning algorithm is introduced. The focus is on the update rules, parameter choices, and theoretical guarantees that are relevant to the empirical comparison.

### 3.1 External-Regret Learners

External-regret learners are used both as standalone baselines and as sublearners inside several swap-regret reductions. The full-information experiments use Hedge, while the bandit experiments use AuerExp3 and Exp3-IX as standalone baselines. Tsallis-INF is used as the bandit sublearner in the Ito reduction.

#### 3.1.1 Hedge

Hedge is the full-information exponential-weights algorithm [5, 9]. Let

$$G_{t-1}(i) = \sum_{s=1}^{t-1} r_s(i)$$

denote the cumulative reward of action  $i$  before round  $t$ . Hedge chooses

$$p_t(i) = \frac{\exp(\eta G_{t-1}(i))}{\sum_{j=1}^K \exp(\eta G_{t-1}(j))}.$$

For a known horizon  $T$ , the implementation uses

$$\eta = \sqrt{\frac{8 \log K}{T}}.$$

For rewards in  $[0, 1]$ , this gives the standard external-regret order

$$O\left(\sqrt{T \log K}\right).$$

The standalone Hedge learner and the Hedge sublearners inside the Blum–Mansour reduction use this fixed-horizon learning rate. The full-information Ito reduction is instead instantiated with anytime Hedge sublearners. Each such sublearner uses

$$\eta_s = \sqrt{\frac{8 \log K}{s}},$$

where  $s$  is the local number of updates received by that sublearner.

### 3.1.2 AuerExp3

AuerExp3 is the explicit-exploration bandit algorithm used both as a standalone baseline and as the sublearner in the bandit Blum–Mansour reduction [2, 10].

Let  $q_t$  denote the exponential-weights distribution before exploration. The distribution from which the action is sampled is

$$p_t(i) = (1 - \gamma)q_t(i) + \frac{\gamma}{K}.$$

After action  $I_t$  is sampled, the reward of action  $i$  is estimated by

$$\hat{r}_t(i) = \frac{r_t(I_t)\mathbf{1}\{I_t = i\}}{p_t(i)}.$$

This importance-weighted estimator allows the exponential-weights update to be performed using only the observed reward.

The experiments use

$$\gamma = \min \left\{ 1, \sqrt{\frac{K \log K}{T}} \right\}, \quad \eta = \frac{\gamma}{K}.$$

These parameters are the ones used in the bandit reduction of Blum and Mansour [2]. The standalone AuerExp3 baseline uses the same parameters so that it is directly comparable with the sublearners used by Blum–Mansour.

### 3.1.3 Exp3-IX

Exp3-IX replaces explicit uniform exploration by implicit exploration [11]. Using losses

$$\ell_t(i) = 1 - r_t(i),$$

the loss estimator is

$$\hat{\ell}_t(i) = \frac{\ell_t(I_t)\mathbf{1}\{I_t = i\}}{p_t(i) + \gamma}.$$

The additional term  $\gamma$  in the denominator controls the variance of the estimator without explicitly mixing the played distribution with the uniform distribution.

The implementation uses

$$\eta = \sqrt{\frac{2 \log K}{KT}}, \quad \gamma = \frac{\eta}{2},$$

following the parameterization of Neu [11]. No additional explicit exploration mixture is applied.

### 3.1.4 Tsallis-INF

Tsallis-INF is an adversarial bandit algorithm based on 1/2-Tsallis entropy regularization [12]. It is not used as a standalone baseline in the experiments, but serves as the external-regret sublearner in the bandit Ito reduction.

The implementation uses the anytime learning-rate schedule

$$\eta_s = \frac{1}{\sqrt{s}},$$

where  $s$  is the local update count of the sublearner, together with the standard importance-weighted loss estimator.

Let  $\hat{L}_s(i)$  denote the cumulative estimated loss of action  $i$ . For numerical computation, define

$$a_i = \eta_s \left( \hat{L}_s(i) - \min_j \hat{L}_s(j) \right).$$

The normalizing value  $z$  is chosen such that

$$\sum_{i=1}^K (a_i + z)^{-2} = 1,$$

which gives the probability distribution

$$p_s(i) = (a_i + z)^{-2}.$$

The scalar normalization equation is solved numerically. This numerical step only computes the distribution defined by the Tsallis-INF update and does not change the learning rule.

Tsallis-INF achieves adversarial bandit regret of order

$$O(\sqrt{Ks}),$$

which will be used below when describing the bandit Ito reduction.

## 3.2 Regret-Matching Procedures

Hart and Mas-Colell introduce regret-matching procedures based on cumulative pairwise replacement gains [1]. The two procedures used in this thesis share the same regret matrix but differ in how the next mixed strategy is obtained.

Let

$$C_t(i, j) = \sum_{s=1}^t \mathbf{1}\{I_s = i\} (r_s(j) - r_s(i))$$

be the cumulative gain from replacing action  $i$  by action  $j$ .

For  $i \neq j$ , define

$$Q_t(i, j) = \frac{[C_t(i, j)]_+}{Kt},$$

where

$$[x]_+ = \max\{x, 0\}.$$

The diagonal entry is chosen so that each row is a probability distribution:

$$Q_t(i, i) = 1 - \sum_{j \neq i} Q_t(i, j).$$

### 3.2.1 Regret Matching

The original regret-matching procedure selects the next strategy using the row of  $Q_t$  associated with the action played in the current round. If  $I_t = i$ , then

$$p_{t+1} = Q_t(i, \cdot).$$

This is the regret-matching procedure studied in Section 2 of Hart and Mas-Colell [1]. Its relevant theoretical guarantee is a joint self-play result: if all players use regret matching, the empirical distribution of play converges almost surely to the set of correlated equilibria.

This guarantee should be distinguished from an individual adversarial no-internal-regret guarantee. In particular, the theorem does not state that a single player using this procedure must achieve vanishing internal regret against an arbitrary adaptive environment. The one-player regret curves for RM are therefore treated as finite-horizon diagnostics rather than as tests of such a guarantee.

### 3.2.2 Stationary Regret Matching

A second procedure in Hart and Mas-Colell [1] uses a stationary distribution of the regret matrix instead of one of its rows. The next strategy satisfies

$$p_{t+1} = p_{t+1} Q_t.$$

This procedure is referred to in this thesis as *Stationary Regret Matching* (SRM). The individual-player result in Hart and Mas-Colell [1] guarantees vanishing pairwise replacement regret without requiring the opponents to follow the same learning rule.

As shown in Chapter 2,

$$R_T^{\text{swap}} \leq K R_T^{\text{int}}.$$

Therefore, for fixed  $K$ , vanishing average internal regret also implies vanishing average swap regret. Although SRM is introduced through internal regret rather than directly as a swap-regret algorithm, this relation makes it relevant to the comparison in this thesis.

## 3.3 Stationary Swap-Regret Reductions

Blum–Mansour, Ito, and LCE-IX use a common stationary-distribution construction. Their main differences lie in the external-regret sublearners, the feedback estimators, and which sublearners are updated at each round.

### 3.3.1 Common Construction

The swap-regret reductions considered in this section maintain  $K$  external-regret sublearners, one for each source action  $i \in [K]$ . Sublearner  $i$  outputs a distribution

$$q_t^{(i)} \in \Delta_K,$$

where choosing action  $j$  can be interpreted as replacing source action  $i$  by action  $j$ .

This construction reflects the structure of a swap deviation. A swap mapping

$$\phi : [K] \rightarrow [K]$$

may choose a different replacement  $\phi(i)$  for every source action  $i$ . The  $K$  sublearners therefore separate this problem into  $K$  external-regret problems: sublearner  $i$  learns which replacement action  $j$  is preferable when the source action is  $i$ .

The sublearner distributions form the rows of a stochastic matrix

$$Q_t = \begin{pmatrix} q_t^{(1)} \\ \vdots \\ q_t^{(K)} \end{pmatrix}.$$

The master strategy  $p_t$  is chosen as a stationary distribution of this matrix:

$$p_t = p_t Q_t.$$

The stationary condition makes the master strategy consistent with the replacement distributions proposed by the sublearners. If source action  $i$  has probability  $p_t(i)$ , then the probability mass transferred from  $i$  to action  $j$  is

$$p_t(i) q_t^{(i)}(j).$$

Summing over all source actions gives

$$\sum_{i=1}^K p_t(i) q_t^{(i)}(j),$$

which, by  $p_t = p_t Q_t$ , is exactly  $p_t(j)$ . Thus, applying the sublearners' replacement distributions to the master strategy reproduces the same master strategy.

Blum–Mansour, Ito, and LCE-IX all use this stationary construction. Their main differences lie in the feedback given to the sublearners and in which sublearners are updated in each round.

The sublearners are chosen to match the corresponding literature-based instantiations and regret guarantees where applicable. The specific choices for each reduction are described below.

### 3.3.2 Blum–Mansour

The Blum–Mansour (BM) reduction updates all  $K$  sublearners in every round [2].

Under full-information feedback, sublearner  $i$  receives the scaled reward vector

$$p_t(i)r_t.$$

The implementation uses fixed-horizon Hedge as the external-regret learner in each row.

With the standard Hedge sublearners used in this thesis, directly summing the per-row Hedge regret bounds gives the conservative order

$$O\left(K\sqrt{T\log K}\right).$$

Blum and Mansour also derive the refined full-information order

$$O\left(\sqrt{TK\log K}\right)$$

with a sharper analysis [2]. The first bound is stated here only as the direct conservative bound for the fixed-Hedge instantiation used in this thesis.

Under bandit feedback, each row instead contains an AuerExp3 learner. Suppose the master action  $I_t$  is selected. Sublearner  $i$  receives the observed gain

$$g_t^{(i)} = \frac{p_t(i)q_t^{(i)}(I_t)}{p_t(I_t)}r_t(I_t).$$

The AuerExp3 learner then applies its own importance weighting with respect to  $q_t^{(i)}(I_t)$ . All  $K$  sublearners are updated in every round.

With the parameter choice used here, the Blum–Mansour analysis gives bandit swap pseudo-regret of order

$$O\left(K\sqrt{KT\log K}\right).$$

### 3.3.3 Ito Reduction

Ito's reduction retains the same stationary matrix construction but changes the update schedule [13]. Instead of updating every sublearner, it first samples an index

$$J_t \sim p_t$$

and then samples the actual action from the selected sublearner:

$$I_t \sim q_t^{(J_t)}.$$

Because  $p_t = p_t Q_t$ , the marginal distribution of  $I_t$  is again  $p_t$ . Only the selected sublearner  $J_t$  is updated in round  $t$ .

If the base learner has a concave external-regret bound  $r(s)$ , Ito's reduction gives

$$\mathbb{E}[R_T^{\text{swap}}] \leq Kr(T/K).$$

In the full-information implementation, the selected sublearner is an anytime Hedge instance. Substituting its external-regret order gives

$$O\left(\sqrt{TK \log K}\right).$$

Under bandit feedback, the selected sublearner is an independent anytime Tsallis-INF instance. Since its external-regret order is

$$O\left(\sqrt{Ks}\right),$$

the resulting swap-regret order is

$$O\left(K\sqrt{T}\right).$$

The local time  $s$  of each sublearner advances only when that sublearner is selected.

### 3.3.4 LCE-IX

LCE-IX is a bandit swap-regret algorithm that combines the stationary reduction with implicit exploration [14]. Like Blum–Mansour, it updates all  $K$  sublearners in every round.

For realized action  $I_t$ , row  $i$  first forms the proportional observed loss

$$Y_t^{(i)} = \frac{p_t(i)q_t^{(i)}(I_t)}{p_t(I_t)}(1 - r_t(I_t)).$$

Its implicit-exploration estimate uses

$$\frac{Y_t^{(i)}}{q_t^{(i)}(I_t) + \gamma_t}.$$

The implementation uses the local-time parameters

$$\eta_t = \sqrt{\frac{\log K}{t}}, \quad \gamma_t = \frac{\eta_t}{2}.$$

The analysis of Huang and Pan [14] gives expected swap regret of order

$$O\left(K\sqrt{T \log K}\right).$$

LCE-IX is therefore included as a bandit-specific alternative to the Blum–Mansour and Ito reductions.

### 3.4 Algorithm Summary

Table 3.1 summarizes the algorithms used in the experiments and their roles in the comparison.

Table 3.1: Algorithms used in the experiments.

| Algorithm | Feedback      | Role in the experiments   |
|-----------|---------------|---------------------------|
| Hedge     | Full          | External-regret baseline  |
| AuerExp3  | Bandit        | External-regret baseline  |
| Exp3-IX   | Bandit        | External-regret baseline  |
| RM        | Full          | Regret-matching benchmark |
| SRM       | Full          | Internal-regret procedure |
| BM        | Full / bandit | Swap-regret reduction     |
| Ito       | Full / bandit | Swap-regret reduction     |
| LCE-IX    | Bandit        | Swap-regret algorithm     |

The guarantees summarized above are not always directly comparable, since they differ in form and probabilistic interpretation.

## 4 Experimental Methodology

This chapter describes the experimental framework used to compare the algorithms introduced in Chapter 3. The experiments cover both one-player time-varying environments and repeated-game self-play under full-information and bandit feedback. All algorithms are evaluated using the same regret definitions, while repeated-game experiments are additionally evaluated through convergence toward correlated and coarse correlated equilibrium.

### 4.1 Experimental Setup

All experiments use rewards in  $[0, 1]$  and a final horizon of

$$T = 10^6.$$

Each configuration is run for eight independent replicates. The base seed is 42, and learner and environment seeds are derived deterministically for each replicate where applicable. This makes the experiments reproducible while keeping the relevant random sources separate.

Full-information experiments compare Hedge, Blum–Mansour (BM), Ito, Regret Matching (RM), and Stationary Regret Matching (SRM). Bandit experiments compare AuerExp3, Exp3-IX, BM, Ito, and LCE-IX. In the repeated-game experiments, both players use the same algorithm, so that each configuration represents self-play.

At every round, a learner samples an action from its current strategy. The environment then determines the rewards, after which the learner receives the feedback allowed by the selected feedback model. Regret evaluation is kept separate from the learner update. In particular, a bandit learner receives only the reward of its sampled action, while the simulator retains the complete reward vector for offline regret computation. Counterfactual rewards used by the evaluator are never passed to the learner.

Regret is evaluated separately for each player in repeated games. Equilibrium evaluation instead uses the complete joint-action history.

The main experiment groups are summarized in Table 4.1.

Table 4.1: Experiment configurations used in the empirical comparison.

| Environment                    | Setting              | Actions      | Feedback      |
|--------------------------------|----------------------|--------------|---------------|
| RPS                            | Two-player self-play | 3 per player | Full / bandit |
| RPSLS                          | Two-player self-play | 5 per player | Full / bandit |
| Historical-frequency adversary | One player           | 9            | Full / bandit |
| Independent lazy random walk   | One player           | 9            | Full / bandit |

## 4.2 Experimental Environments

The experiments use two repeated games and two one-player environments. The repeated games provide a setting in which regret can be related to equilibrium convergence, while the one-player environments isolate the regret-minimization behavior of the algorithms.

### 4.2.1 Repeated Games

**Rock–Paper–Scissors.** Rock–Paper–Scissors (RPS) is a symmetric constant-sum game with three actions. Payoffs are normalized as

$$\text{win} = 1, \quad \text{tie} = 0.5, \quad \text{loss} = 0.$$

RPS provides the smallest cyclic game used in the experiments and serves as a simple benchmark for comparing self-play dynamics.

**Rock–Paper–Scissors–Lizard–Spock.** RPSLS extends the same cyclic structure to five actions: Rock, Paper, Scissors, Lizard, and Spock. Each action beats two actions and loses to two, with the same normalized payoff convention as RPS. The game follows the five-action cyclic example used in the self-play setting of Paes Leme et al. [15].

Compared with RPS, RPSLS preserves a similar strategic structure while increasing the number of available deviations. It therefore provides a second test of both regret and equilibrium convergence in self-play.

### 4.2.2 Historical-Frequency Adversary

The historical-frequency adversary is an adaptive one-player environment. Before the current action is added to the history, the

$$\left\lceil \frac{K}{2} \right\rceil$$

most frequently selected actions receive reward 0, while all remaining actions receive reward 1. For the experiments with  $K = 9$ , five actions are therefore punished at every round.

Ties between equally frequent actions are resolved by a deterministic rotating rule. The resulting reward vector depends on the learner's previous actions, but not on its fresh random draw in the current round. The environment is therefore adaptive while remaining non-anticipating.

This construction is intended as a stress test for action-dependent regret. Actions that are repeatedly favored by the learner are eventually punished, which can expose differences between algorithms that control only fixed deviations and algorithms designed to control internal or swap deviations.

### 4.2.3 Independent Lazy Random Walk

The independent lazy random walk provides a contrasting one-player environment whose reward process does not depend on the learner's actions. Each action follows an independent random walk on

$$\{0, 0.1, \dots, 0.9, 1\},$$

and all actions start with reward 0.5.

At an interior state, the reward moves down by 0.1, remains unchanged, or moves up by 0.1, each with probability  $1/3$ . At either boundary, the process remains at the boundary or moves one step inward, each with probability  $1/2$ .

The complete reward trajectories are generated from the environment seed independently of the learner's actions. Within each replicate, the same environment realization is used across algorithms. This allows their performance to be compared on the same underlying reward sequence.

## 4.3 Regret Evaluation

The theoretical guarantees described in Chapter 3 are not always directly comparable, since they differ in form and probabilistic interpretation. The experiments therefore evaluate every algorithm using the same external-, internal-, and swap-regret measures introduced in Chapter 2.

### 4.3.1 Common Regret Computation

All three regret notions are computed from cumulative replacement gains.

For realized regret, define

$$G_T(i, j) = \sum_{t=1}^T \mathbf{1}\{I_t = i\} (r_t(j) - r_t(i)).$$

For strategy-weighted regret, define analogously

$$\bar{G}_T(i, j) = \sum_{t=1}^T p_t(i) (r_t(j) - r_t(i)).$$

For the realized replacement-gain matrix  $G_T$ , external regret is

$$R_T^{\text{ext}} = \max_j \sum_i G_T(i, j),$$

internal regret is

$$R_T^{\text{int}} = \max_{i,j} G_T(i, j),$$

and swap regret is

$$R_T^{\text{swap}} = \sum_i \max_j G_T(i, j).$$

Applying the same formulas to  $\bar{G}_T$  gives the corresponding strategy-weighted regret quantities.

Thus, the algorithms are not evaluated only under the regret notion for which they were designed. The same evaluator measures all three deviation classes for every learner.

Full-information results use strategy-weighted regret. For consistency with the implementation, this quantity is labeled *expected regret* in the figures. Bandit results use realized regret. The choice affects only the evaluation and does not alter the information supplied to the learner.

### 4.3.2 Convergence Measures and Statistical Reporting

The experiments report both average regret,

$$\frac{R_t}{t},$$

and the normalized quantity

$$\frac{R_t}{\sqrt{t}}.$$

The former directly reflects the no-regret criterion  $R_t/t \rightarrow 0$ , while the latter is the main diagnostic for comparing finite-horizon convergence behavior. Since many of the theoretical guarantees in Chapter 3 have  $\sqrt{t}$ -type dependence, a bounded  $R_t/\sqrt{t}$  trajectory is consistent with such regret growth and makes differences between algorithms more visible than  $R_t/t$ .

For algorithms with fixed-horizon parameters, intermediate points of a run with final horizon  $T = 10^6$  are not interpreted as separately tuned experiments at those shorter horizons. All curves are computed separately for eight replicates and reported as replicate means with pointwise two-sided 95% Student- $t$  confidence intervals.

## 4.4 Equilibrium Evaluation

Equilibrium convergence is evaluated in the RPS and RPSLS self-play experiments. From the joint-action history, the empirical distribution after  $t$  rounds is

$$\hat{\mu}_t(a) = \frac{1}{t} \sum_{s=1}^t \mathbf{1}\{a_s = a\}.$$

The experiments measure the  $L_1$  distance of this empirical distribution to the complete correlated-equilibrium and coarse-correlated-equilibrium sets:

$$d_{\text{CE}}(\hat{\mu}_t) = \min_{\mu \in \text{CE}} \|\hat{\mu}_t - \mu\|_1,$$

and

$$d_{\text{CCE}}(\hat{\mu}_t) = \min_{\mu \in \text{CCE}} \|\hat{\mu}_t - \mu\|_1.$$

Since

$$\text{CE} \subseteq \text{CCE},$$

we have

$$d_{\text{CCE}}(\hat{\mu}_t) \leq d_{\text{CE}}(\hat{\mu}_t).$$

The distances are computed in the full joint-action probability space by solving the corresponding linear programs. They are evaluated separately for each replicate, and the figures report the replicate mean together with pointwise 95% Student- $t$  confidence intervals.

## 5 Experimental Results

This chapter presents the empirical comparison of the algorithms introduced in Chapter 3. The results are divided into full-information and bandit feedback because the two settings compare different sets of five algorithms. Within each setting, the focus is on differences between the algorithms across the four experimental environments rather than on comparing the environments themselves.

For each regret experiment, the left column reports average regret  $R_t/t$ , while the right column reports  $R_t/\sqrt{t}$ . The former directly reflects the no-regret criterion, while the latter provides a clearer finite-horizon comparison of regret growth. The three rows correspond to external, internal, and swap regret.

For the symmetric self-play experiments, only Player 0 is shown in the regret figures to avoid duplicating the same comparison for both players. Equilibrium distances are computed from the complete joint-action history and therefore depend on the behavior of both players. All figures report the mean over eight replicates with pointwise 95% Student- $t$  confidence intervals.

### 5.1 Full-Information Comparison

The full-information experiments compare Hedge, Regret Matching (RM), Stationary Regret Matching (SRM), Blum–Mansour (BM), and Ito. Hedge serves as the external-regret baseline, BM and Ito are swap-regret reductions, and SRM provides an internal-regret procedure. RM is included separately because its main theoretical guarantee concerns joint self-play convergence [1].

#### 5.1.1 Regret Comparison

Figures 5.1–5.4 show the regret trajectories in the two one-player environments and the two repeated games.

**Hedge.** Across all four full-information experiments, Hedge achieves low external regret but substantially larger internal and swap regret. This is consistent with the standard theoretical guarantee for Hedge, which controls external regret but does not provide a corresponding guarantee for internal or swap deviations [5, 9].

The experiments therefore reproduce the theoretical distinction between fixed-action deviations and the stronger action-dependent deviation classes.

## 5 Experimental Results

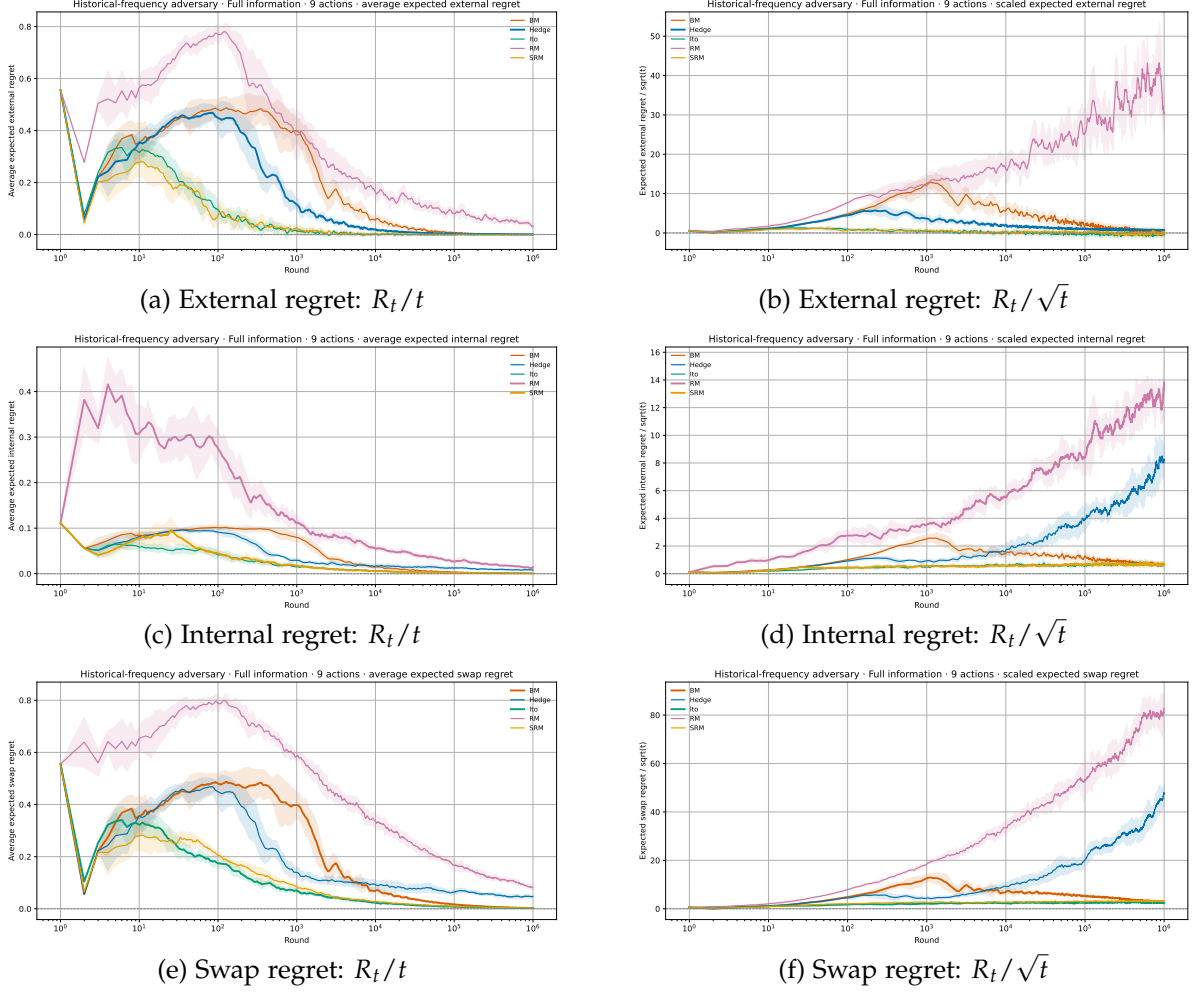


Figure 5.1: Regret comparison under full-information feedback in the historical-frequency adversary,  $K = 9$ ,  $T = 10^6$ , eight replicates.

## 5 Experimental Results

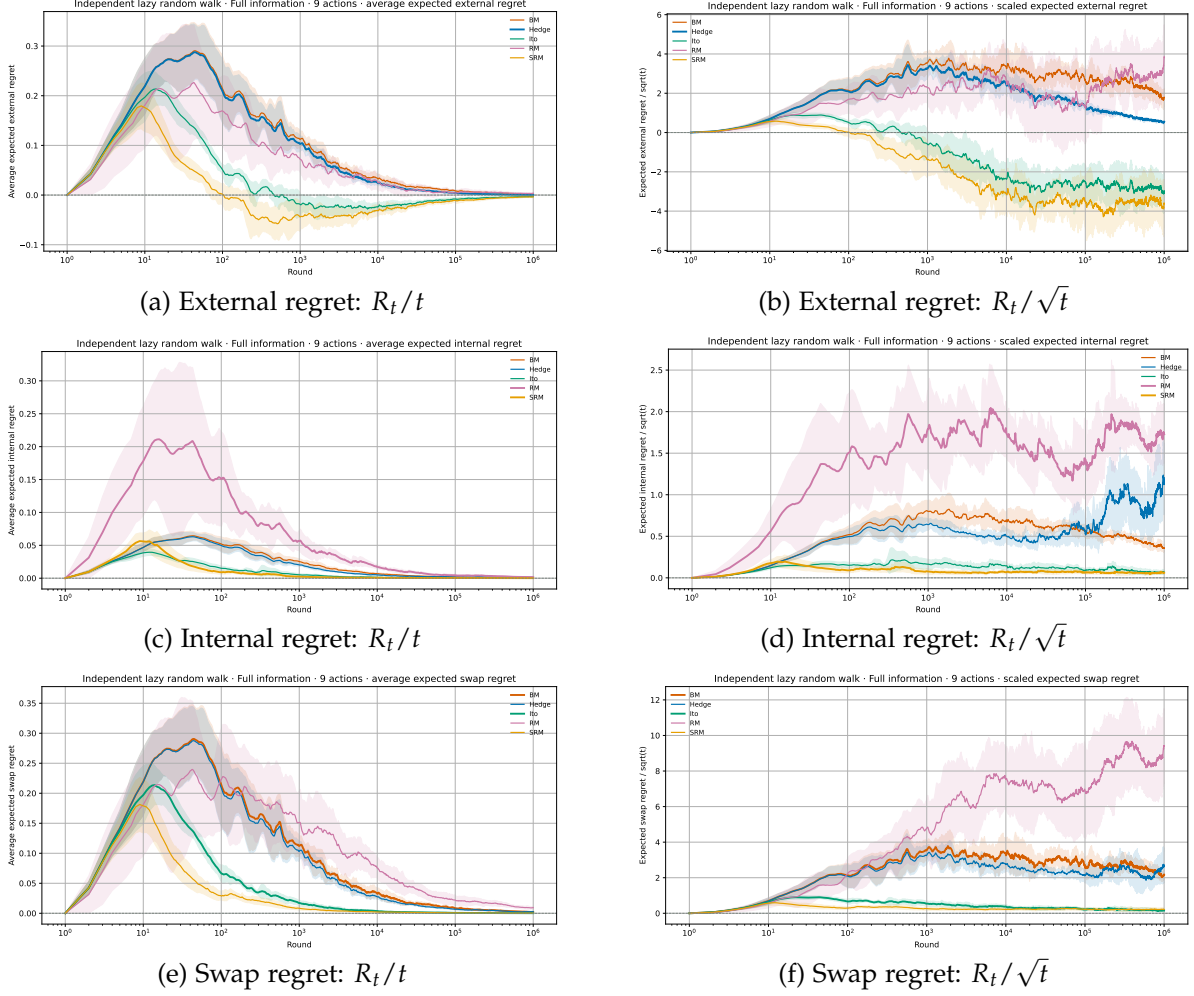
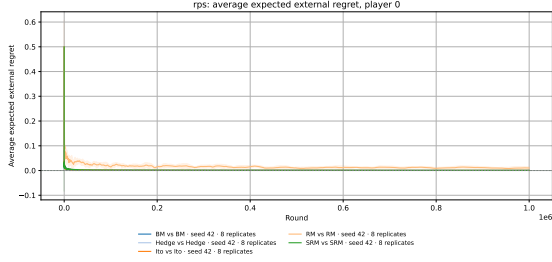
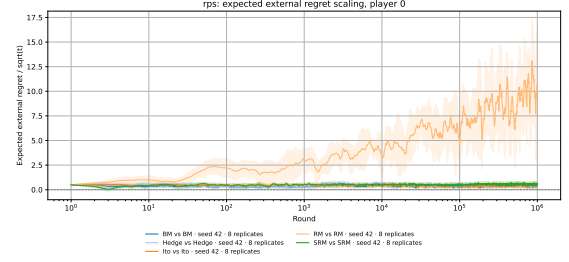


Figure 5.2: Regret comparison under full-information feedback in the independent lazy random walk,  $K = 9$ ,  $T = 10^6$ , eight replicates.

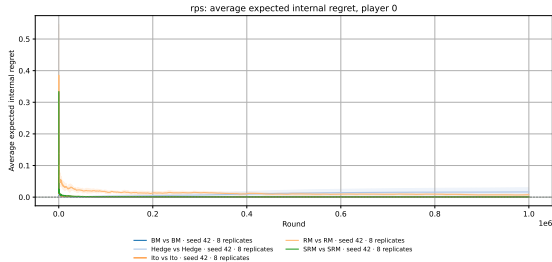
## 5 Experimental Results



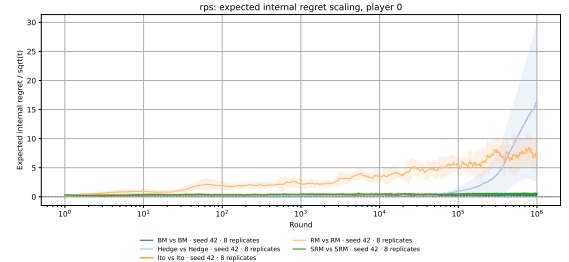
(a) External regret:  $R_t/t$



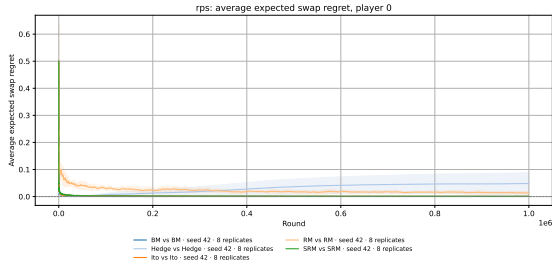
(b) External regret:  $R_t/\sqrt{t}$



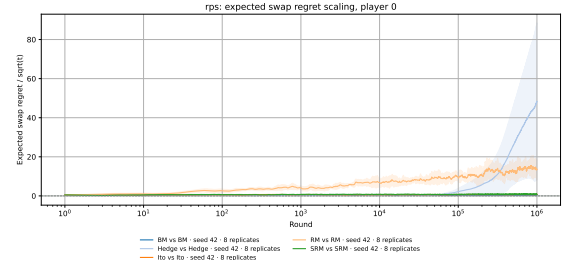
(c) Internal regret:  $R_t/t$



(d) Internal regret:  $R_t/\sqrt{t}$



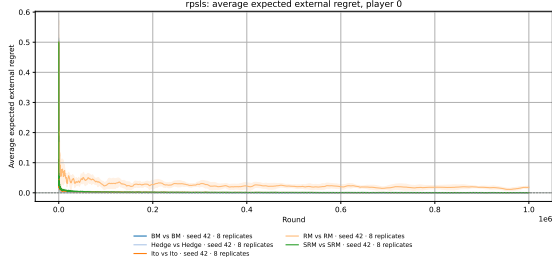
(e) Swap regret:  $R_t/t$



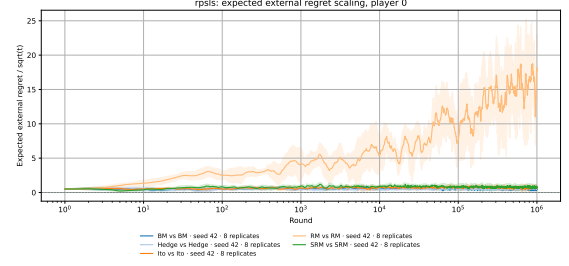
(f) Swap regret:  $R_t/\sqrt{t}$

Figure 5.3: Player 0 regret in RPS self-play under full-information feedback,  $T = 10^6$ , eight replicates.

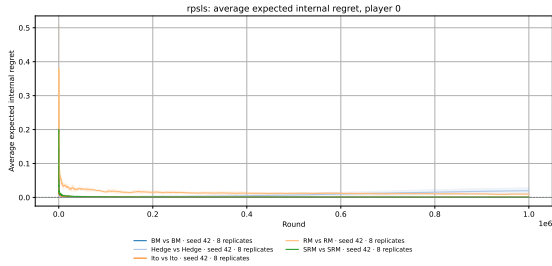
## 5 Experimental Results



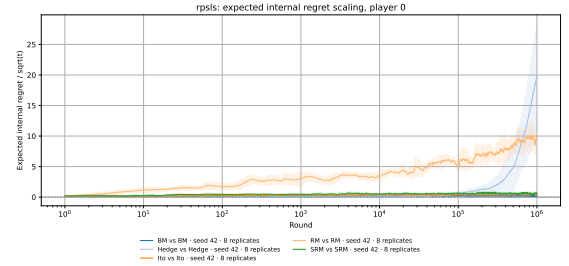
(a) External regret:  $R_t/t$



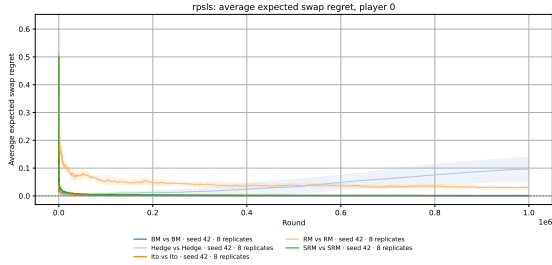
(b) External regret:  $R_t/\sqrt{t}$



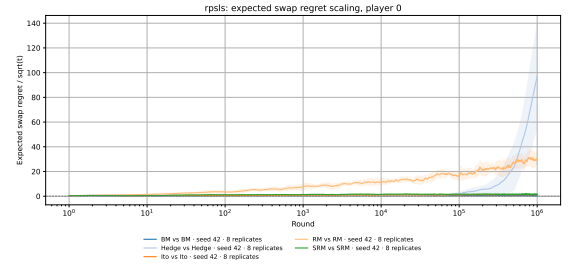
(c) Internal regret:  $R_t/t$



(d) Internal regret:  $R_t/\sqrt{t}$



(e) Swap regret:  $R_t/t$



(f) Swap regret:  $R_t/\sqrt{t}$

Figure 5.4: Player 0 regret in RPSLS self-play under full-information feedback,  $T = 10^6$ , eight replicates.

**Blum–Mansour and Ito.** BM and Ito provide a particularly direct comparison with standalone Hedge because their full-information implementations use Hedge as the external-regret sublearner. Nevertheless, both reductions achieve substantially lower swap regret than Hedge across the four experiments.

This qualitative difference is consistent with the purpose of the two reductions: BM converts external-regret sublearners into a swap-regret procedure [2], while Ito provides an alternative swap-regret reduction based on selectively updating the sublearners [13].

Beyond this agreement with the theoretical guarantees, the experiments reveal a clear finite-horizon difference between the two reductions. Ito generally reaches low swap regret substantially earlier than BM in the full-information experiments.

**Regret Matching.** RM behaves differently from the other procedures. Its reported strategy-weighted external regret does not exhibit the same vanishing behavior as Hedge, BM, Ito, or SRM, and its internal and swap regret can also remain comparatively large.

This observation does not contradict the classical regret-matching result. The procedure used here is the regret-matching algorithm whose relevant guarantee in Hart and Mas-Colell [1] concerns the empirical distribution generated when all players use the procedure in self-play. It is not an individual adversarial no-internal-regret guarantee. The corresponding self-play behavior is examined below.

**Stationary Regret Matching.** SRM shows consistently strong finite-horizon performance across all four experiments. The individual-player result for the stationary procedure in Hart and Mas-Colell [1] guarantees vanishing pairwise replacement regret. For the fixed finite action sets considered here,

$$R_T^{\text{swap}} \leq KR_T^{\text{int}},$$

so vanishing average internal regret also implies vanishing average swap regret.

The empirical results agree with this prediction: SRM achieves low internal and swap regret in both one-player environments and both self-play games. Beyond the qualitative theoretical guarantee, SRM is also among the fastest-converging full-information procedures in several of the experiments.

### 5.1.2 Equilibrium Convergence

Figures 5.5 and 5.6 compare the empirical self-play distributions with the CE and CCE sets.

The equilibrium results closely follow the corresponding theoretical distinction between the regret notions. Hedge self-play approaches the CCE set in both games while remaining noticeably farther from the CE set. This is consistent with the standard implication from vanishing external regret to coarse correlated equilibrium [3].

BM and Ito instead approach the CE set much more closely. Their behavior is consistent with the fact that vanishing swap regret implies vanishing internal regret and therefore convergence of empirical play toward correlated equilibrium [2, 4].

## 5 Experimental Results

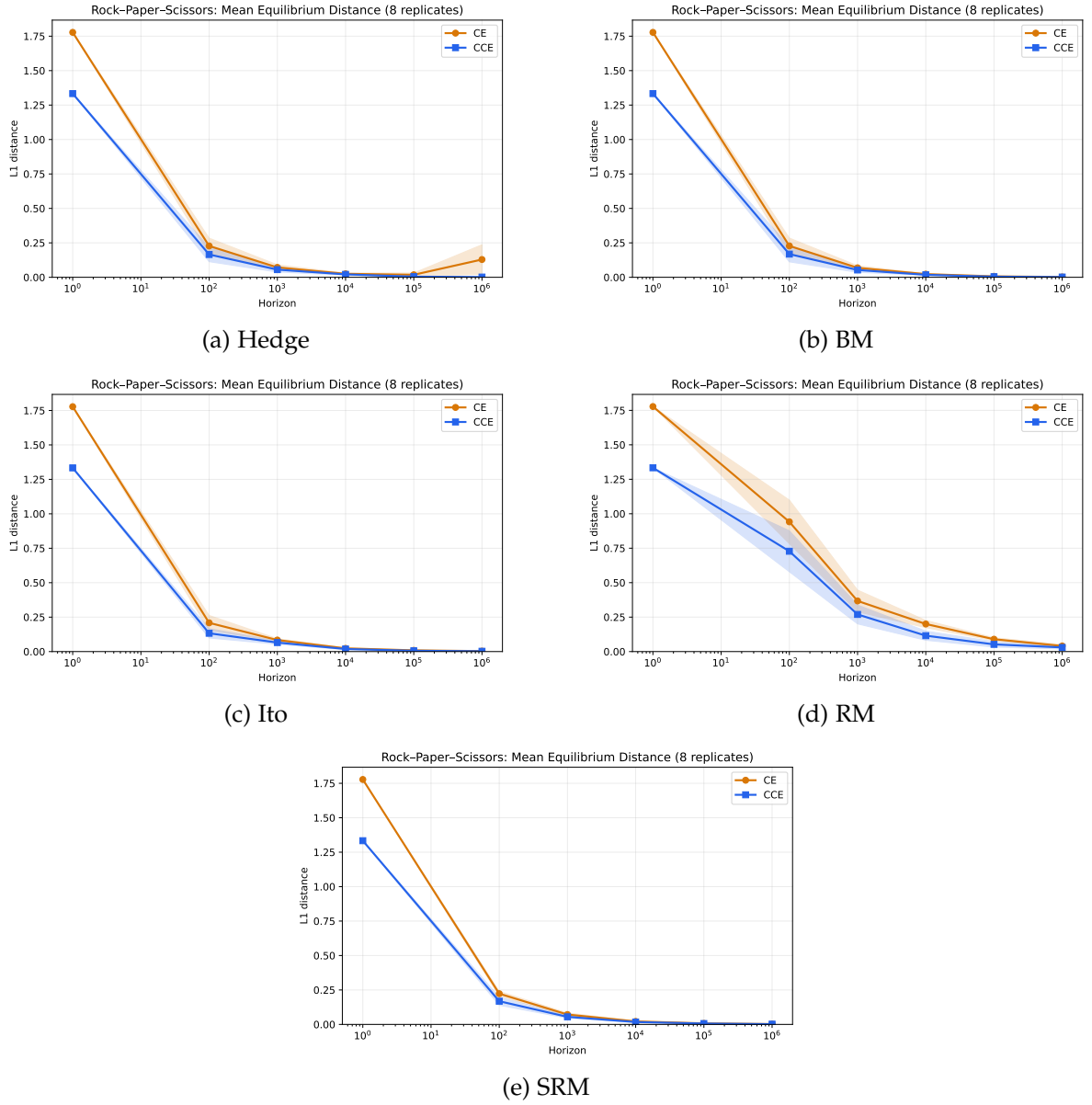
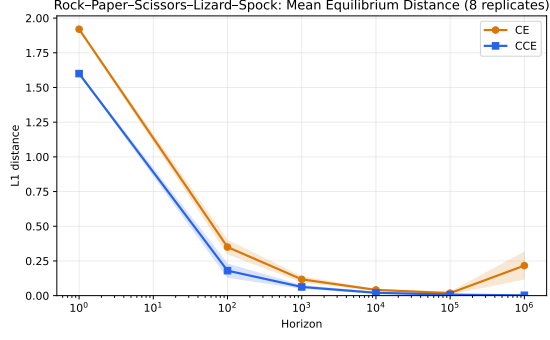
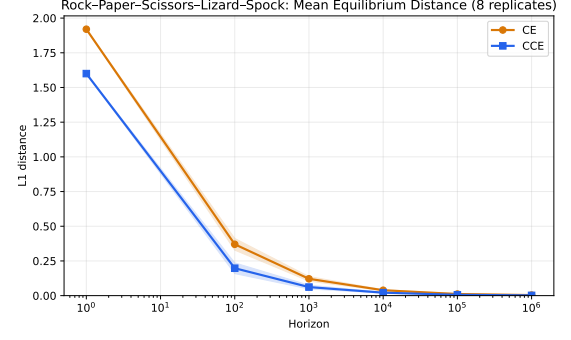


Figure 5.5: Mean  $L_1$  distance of the empirical joint-action distribution to the CE and CCE sets in RPS self-play under full-information feedback.

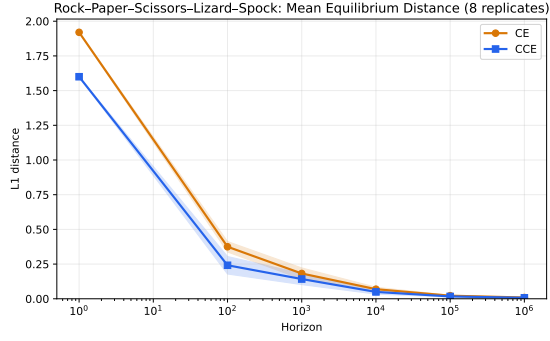
## 5 Experimental Results



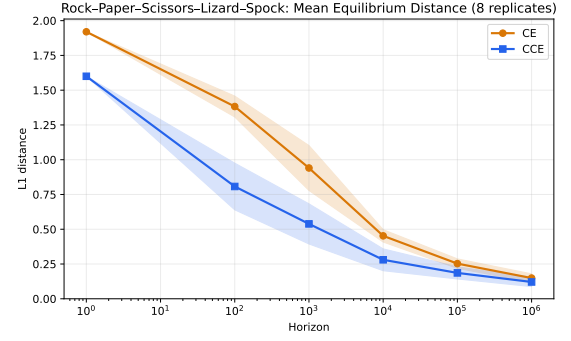
(a) Hedge



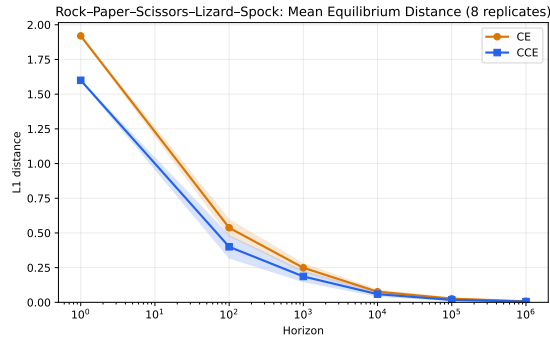
(b) BM



(c) Ito



(d) RM



(e) SRM

Figure 5.6: Mean  $L_1$  distance of the empirical joint-action distribution to the CE and CCE sets in RPSLS self-play under full-information feedback.

This comparison is especially direct because standalone Hedge and the Hedge-based BM and Ito reductions share the same type of external-regret sublearner. The experiments show that the stronger regret control produced by the reductions is accompanied by the stronger equilibrium behavior predicted by the theory.

SRM also approaches CE rapidly in both repeated games. This agrees with the connection between vanishing internal regret and correlated equilibrium [1, 4].

RM gives a different but equally theory-consistent result. Although its reported strategy-weighted regret can remain comparatively large, its realized self-play distribution approaches the CE set in both RPS and RPSLS. This is exactly the type of joint self-play behavior established for regret matching by Hart and Mas-Colell [1].

The RM experiments therefore illustrate that convergence of the realized joint-action distribution toward CE can coexist with large individual strategy-weighted external-regret diagnostics. This does not contradict the regret–equilibrium connection discussed in Chapter 2: the theoretical implication runs from suitable regret control to equilibrium convergence, whereas the converse need not hold. In addition, the equilibrium distance is computed from realized joint play, while the regret figures report strategy-weighted regret.

## 5.2 Bandit Comparison

The bandit experiments compare AuerExp3, Exp3-IX, BM, Ito, and LCE-IX. AuerExp3 and Exp3-IX serve as external-regret baselines [10, 11], while BM, Ito, and LCE-IX are included for their bandit swap-regret guarantees [2, 13, 14].

### 5.2.1 Regret Comparison

Figures 5.7–5.10 show the regret trajectories for the five bandit algorithms.

Across the bandit experiments, the main qualitative distinction is between the external-regret baselines AuerExp3 and Exp3-IX and the swap-regret methods BM, Ito, and LCE-IX. This separation is most pronounced under the historical-frequency adversary and is also visible in the two self-play games. AuerExp3 and Exp3-IX achieve low external regret but retain larger internal and swap regret, whereas BM, Ito, and LCE-IX show substantially stronger control of the action-dependent regret measures. This is consistent with their respective theoretical guarantees [2, 10, 11, 13, 14].

The independent lazy random walk is the main exception. In this environment, the five algorithms behave much more similarly, and even AuerExp3 and Exp3-IX exhibit low observed swap regret over the tested horizon. This does not conflict with the theoretical distinction between the algorithms: the absence of a worst-case swap-regret guarantee does not imply large swap regret on every particular reward sequence.

Among the three swap-regret methods, the experiments do not reveal a stable finite-horizon ordering between BM and Ito; each performs better in some configurations. LCE-IX shows particularly consistent regret performance across the tested bandit settings. These relative performance differences are empirical observations and are not implied by the asymptotic regret guarantees [2, 13, 14].

### 5.2.2 Equilibrium Convergence

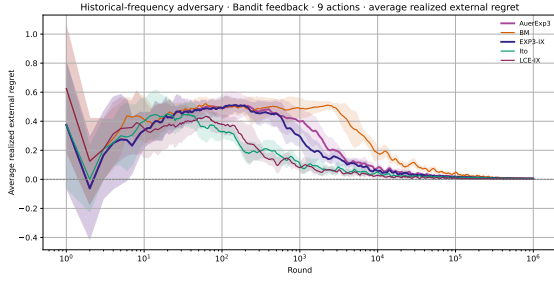
Figures 5.11 and 5.12 show how the differences in regret behavior are reflected in the empirical self-play distributions.

The equilibrium-distance results are less sharply separated than the regret curves. AuerExp3 and Exp3-IX can also reach small CE distance in these games, which is not ruled out by their external-regret guarantees. Their behavior remains consistent with the standard connection between vanishing external regret and coarse correlated equilibrium [3].

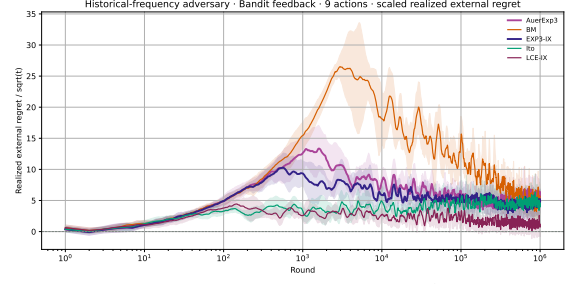
BM, Ito, and LCE-IX likewise approach the CE set closely, consistent with the implication from swap-regret control to internal-regret control and hence to correlated-equilibrium convergence [2, 4, 14].

Within the three swap-regret methods, the equilibrium-distance results do not show a clear and consistent ordering. BM, Ito, and LCE-IX all achieve small CE distances, but none dominates uniformly across both self-play games. Thus, the particularly consistent regret performance of LCE-IX does not translate into an equally clear advantage in equilibrium distance.

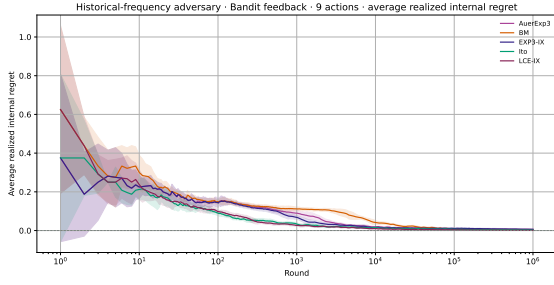
## 5 Experimental Results



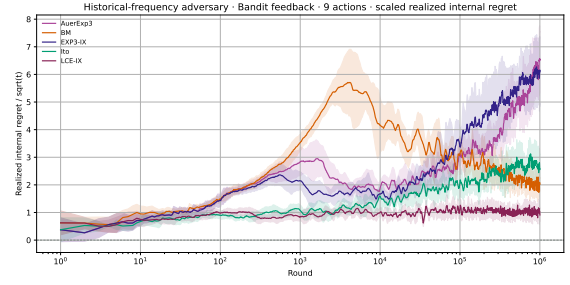
(a) External regret:  $R_t/t$



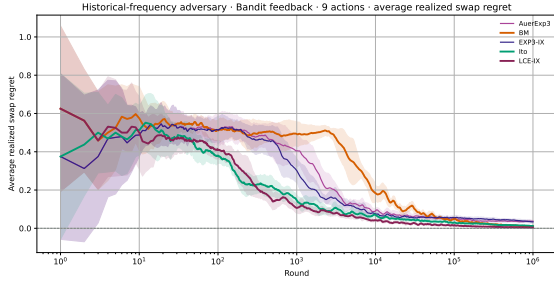
(b) External regret:  $R_t/\sqrt{t}$



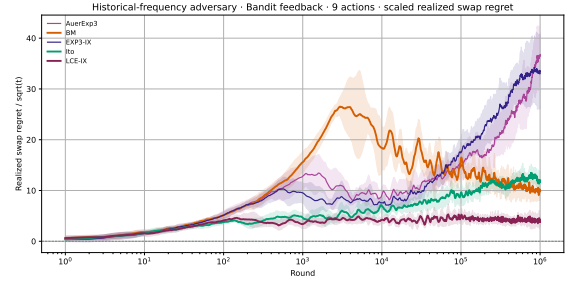
(c) Internal regret:  $R_t/t$



(d) Internal regret:  $R_t/\sqrt{t}$



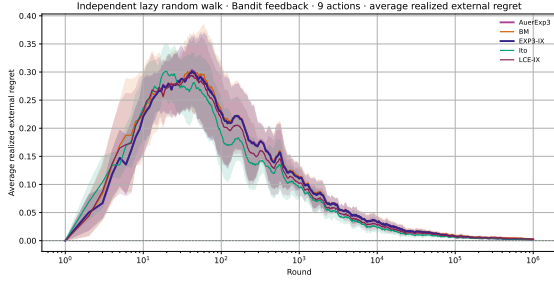
(e) Swap regret:  $R_t/t$



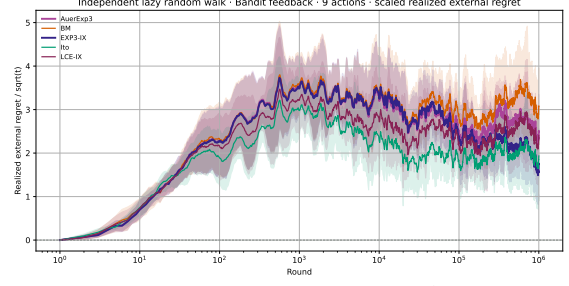
(f) Swap regret:  $R_t/\sqrt{t}$

Figure 5.7: Regret comparison under bandit feedback in the historical-frequency adversary,  $K = 9$ ,  $T = 10^6$ , eight replicates.

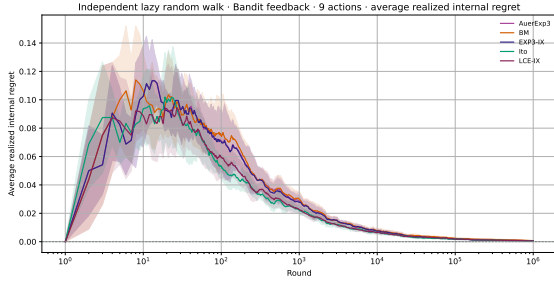
## 5 Experimental Results



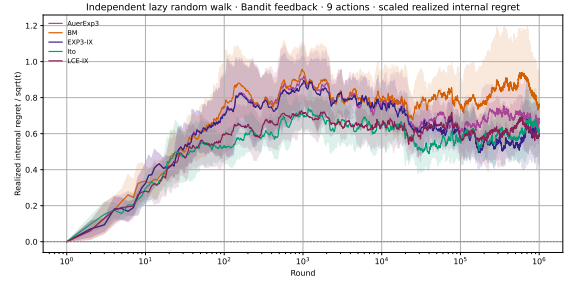
(a) External regret:  $R_t/t$



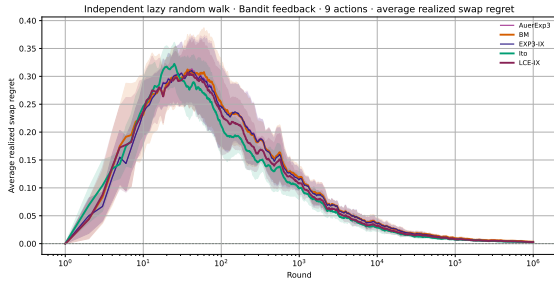
(b) External regret:  $R_t/\sqrt{t}$



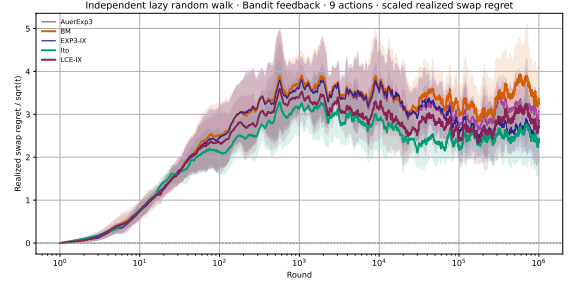
(c) Internal regret:  $R_t/t$



(d) Internal regret:  $R_t/\sqrt{t}$



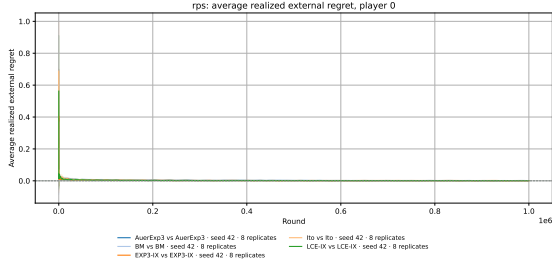
(e) Swap regret:  $R_t/t$



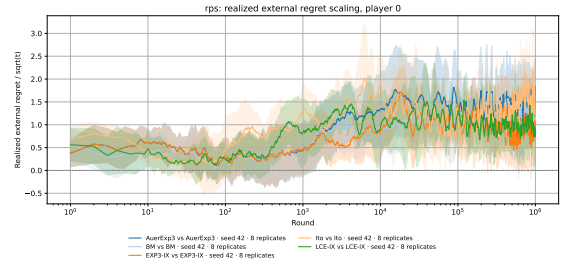
(f) Swap regret:  $R_t/\sqrt{t}$

Figure 5.8: Regret comparison under bandit feedback in the independent lazy random walk,  $K = 9$ ,  $T = 10^6$ , eight replicates.

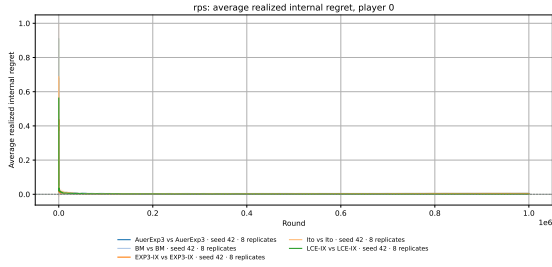
## 5 Experimental Results



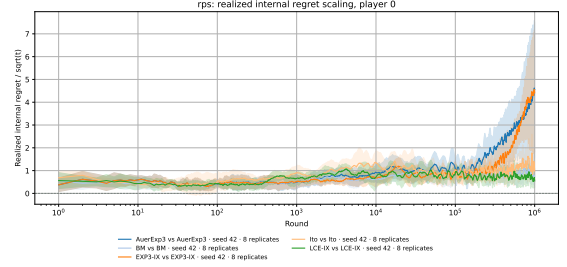
(a) External regret:  $R_t/t$



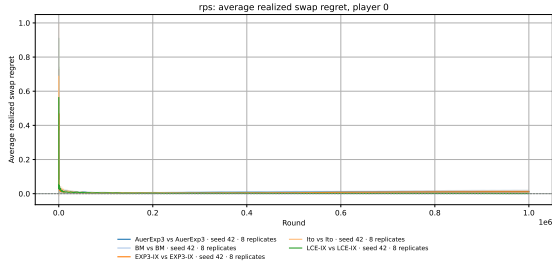
(b) External regret:  $R_t/\sqrt{t}$



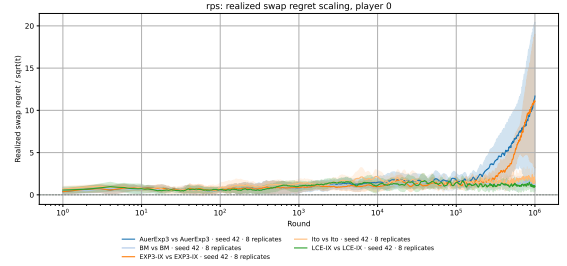
(c) Internal regret:  $R_t/t$



(d) Internal regret:  $R_t/\sqrt{t}$



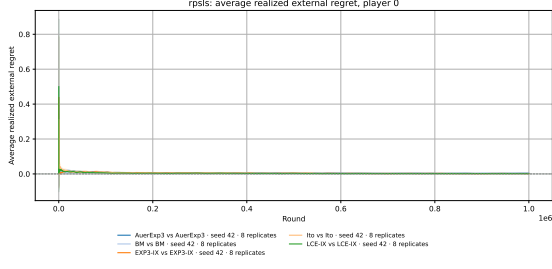
(e) Swap regret:  $R_t/t$



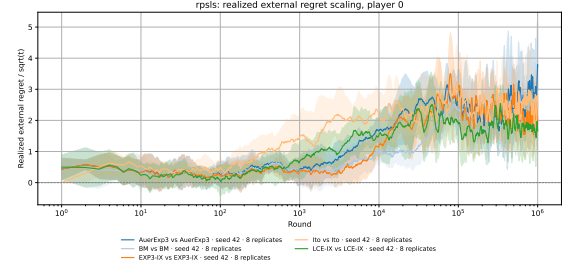
(f) Swap regret:  $R_t/\sqrt{t}$

Figure 5.9: Player 0 regret in RPS self-play under bandit feedback,  $T = 10^6$ , eight replicates.

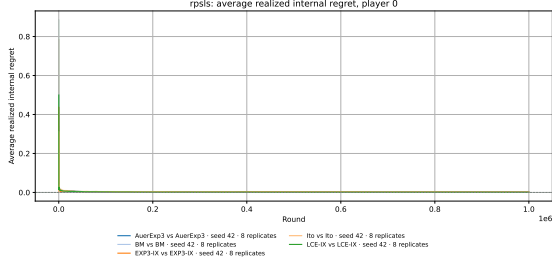
## 5 Experimental Results



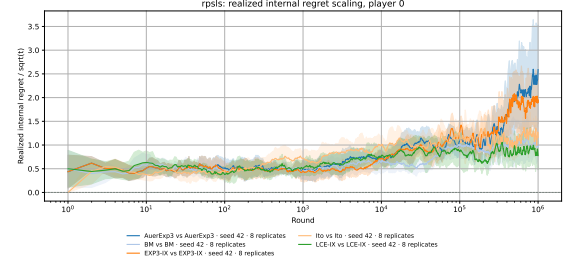
(a) External regret:  $R_t/t$



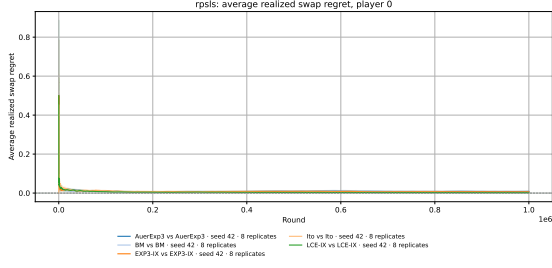
(b) External regret:  $R_t/\sqrt{t}$



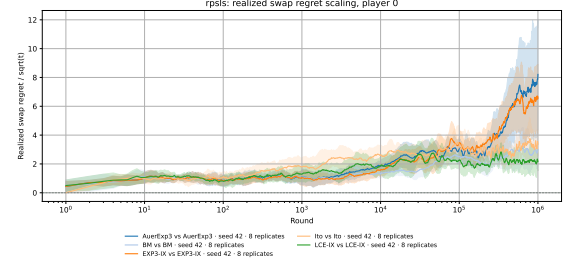
(c) Internal regret:  $R_t/t$



(d) Internal regret:  $R_t/\sqrt{t}$



(e) Swap regret:  $R_t/t$



(f) Swap regret:  $R_t/\sqrt{t}$

Figure 5.10: Player 0 regret in RPSLS self-play under bandit feedback,  $T = 10^6$ , eight replicates.

## 5 Experimental Results

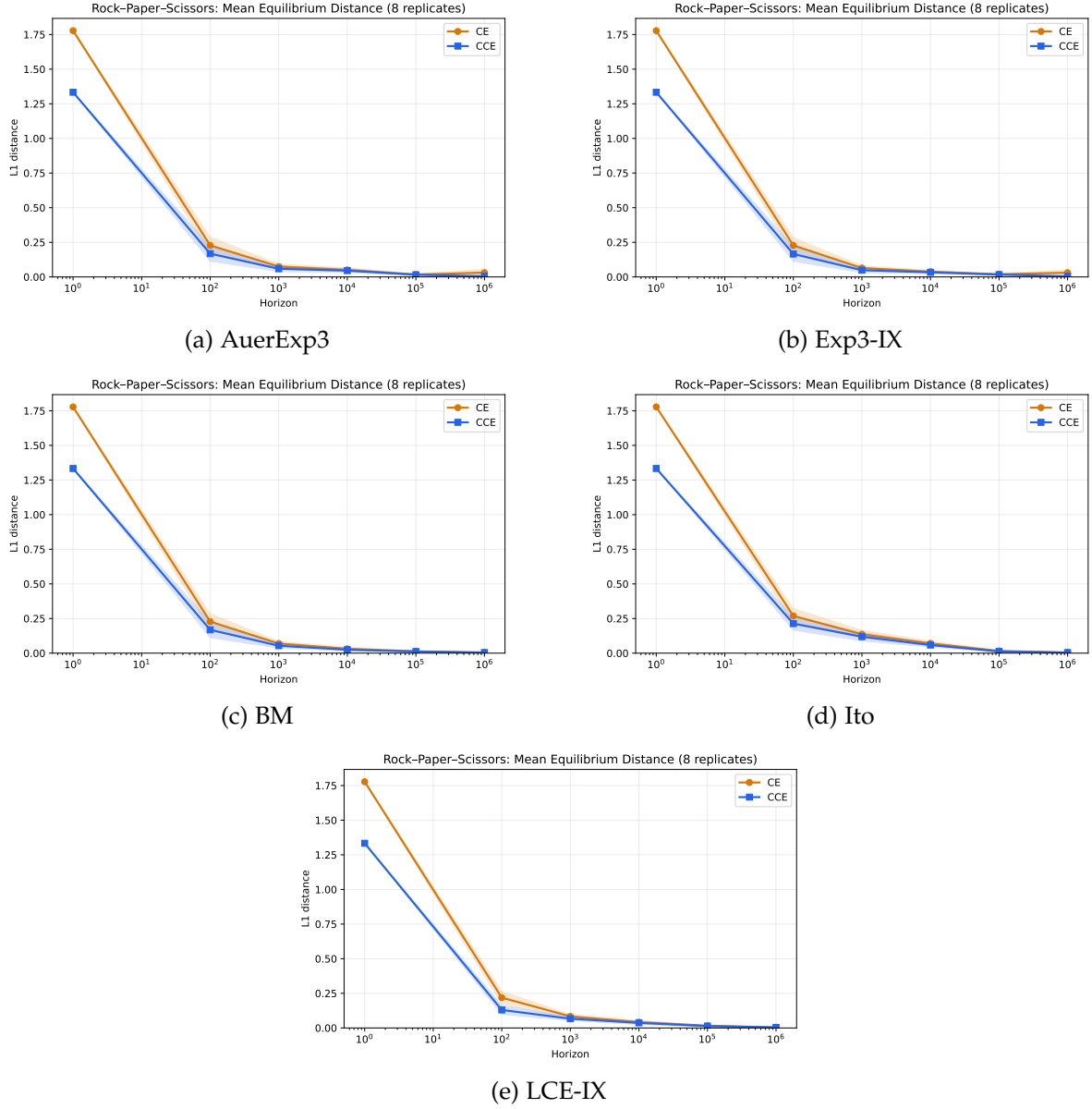
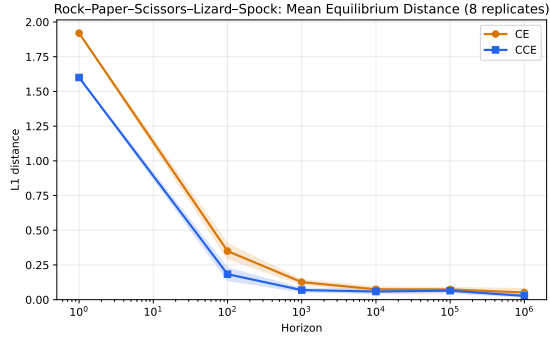
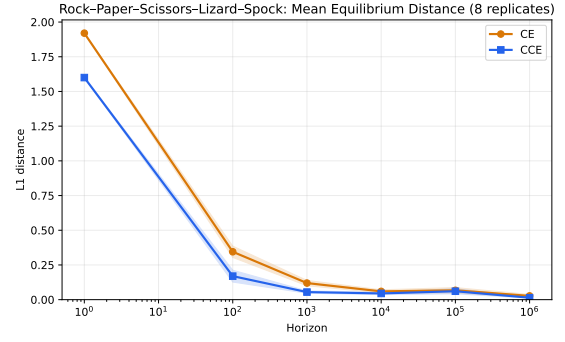


Figure 5.11: Mean  $L_1$  distance of the empirical joint-action distribution to the CE and CCE sets in RPS self-play under bandit feedback.

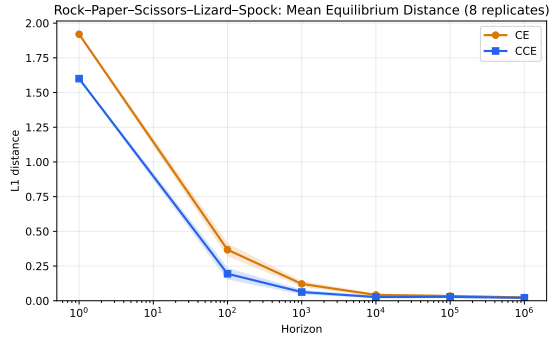
## 5 Experimental Results



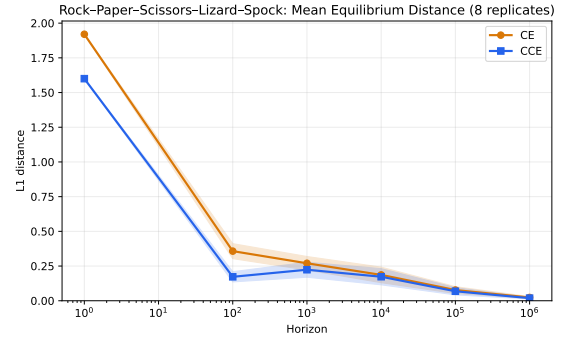
(a) AuerExp3



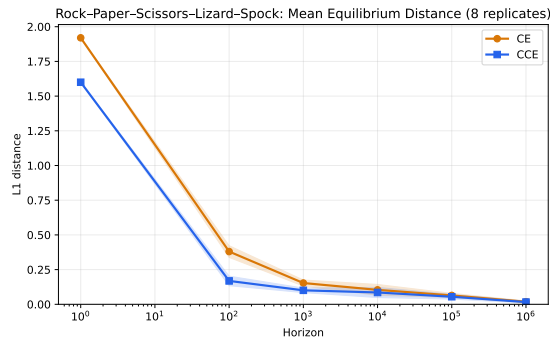
(b) Exp3-IX



(c) BM



(d) Ito



(e) LCE-IX

Figure 5.12: Mean  $L_1$  distance of the empirical joint-action distribution to the CE and CCE sets in RPSLS self-play under bandit feedback.

### 5.3 Summary of Empirical Findings

The experimental results are largely consistent with the theoretical classification of the algorithms, while also revealing finite-horizon differences that are not determined by the guarantees alone.

Under full-information feedback, Hedge consistently achieves low external regret without comparable internal- or swap-regret control, in line with its theoretical role as an external-regret learner [5, 9]. In self-play, Hedge approaches CCE more closely than CE, consistent with the standard external-regret–CCE connection [3].

BM and Ito, although instantiated with Hedge-based sublearners, achieve much stronger swap-regret control and correspondingly approach CE. Ito generally reaches low swap regret earlier than BM in the full-information experiments. SRM also performs strongly, with rapid internal-, swap-, and CE convergence, consistent with its pairwise replacement-regret guarantee [1].

RM exhibits a different pattern: its strategy-weighted regret diagnostics can remain comparatively large, while its realized self-play distribution still approaches CE. This agrees with the joint self-play result of Hart and Mas-Colell [1] and illustrates that regret-to-equilibrium implications need not hold in reverse.

Under bandit feedback, AuerExp3 and Exp3-IX are clearly separated from BM, Ito, and LCE-IX in the historical-frequency and self-play experiments. The external-regret baselines retain larger internal and swap regret, whereas the three swap-regret methods provide stronger action-dependent regret control, consistent with their respective guarantees [2, 10, 11, 13, 14].

The independent lazy random walk is the main exception: all five algorithms behave much more similarly, and even the external-regret baselines exhibit low observed swap regret over the tested horizon.

Among the bandit swap-regret methods, BM and Ito show no stable finite-horizon ordering. LCE-IX has particularly consistent regret performance, but this does not translate into a clear advantage in equilibrium distance, where BM, Ito, and LCE-IX all show strong CE convergence.

Overall, the experiments reproduce the main qualitative distinctions predicted by regret and equilibrium theory, while showing that finite-horizon convergence speed and the ranking among stronger-regret algorithms remain instance-dependent.

## 6 Discussion

The experiments show that the theoretical regret guarantees explain several important differences between the algorithms, but they do not determine finite-horizon performance completely. Some of the observed behavior can be related directly to the structure of the algorithms, while other differences remain empirical observations.

### 6.1 Regret Classes and Algorithm Structure

The most basic difference between the algorithms is the class of deviations they control. External regret only compares the learner with one fixed action. Internal and swap regret allow the alternative action to depend on the action that was originally played.

This difference is also reflected in the algorithm design. BM and Ito separate the learning problem by source action and can therefore react to deviations of the form  $i \mapsto j$ . Standalone external-regret learners do not directly control such action-dependent replacements [2, 13].

This also explains why a stronger regret guarantee does not necessarily lead to a large advantage in every experiment. If a reward sequence creates few profitable action-dependent replacements, an external-regret learner may also show low internal or swap regret. The stronger guarantee mainly becomes important when such replacements occur frequently.

### 6.2 Full-Information Algorithms

The strong performance of SRM also has a natural structural explanation. SRM is built directly from positive pairwise replacement regrets, which are the quantities relevant for internal regret. Its strategy therefore reacts directly to the deviations it aims to reduce, rather than obtaining stronger regret through an external-regret reduction. This may help explain its fast internal-regret convergence in the tested settings [1].

RM has to be interpreted differently. Its classical result concerns the joint behavior of players who all use regret matching: their empirical distribution of play converges toward correlated equilibrium [1]. Its good CE convergence in self-play therefore does not require the individual strategy-weighted regret diagnostics to behave like those of SRM or the swap-regret reductions. The two observations are compatible with the scope of the theoretical result.

### 6.3 Environment-Dependent Algorithm Separation

The difference between external- and swap-regret algorithms is much clearer under the historical-frequency adversary than under the independent lazy random walk. A plausible explanation is that the two environments create very different opportunities for action-dependent deviations.

The historical-frequency adversary reacts to the learner's past action frequencies. Frequently played actions are later assigned lower rewards, so replacing one previously played action by another can become profitable. This naturally creates the type of conditional deviation measured by internal and swap regret.

The independent lazy random walk has a different structure. Its rewards evolve independently of the learner and change only gradually. Favorable actions can remain favorable for relatively long periods, so simply tracking good actions may already avoid many profitable replacements. This could explain why even the external-regret algorithms show low observed swap regret in this environment.

This interpretation remains a hypothesis. The experiments do not directly measure how often profitable pairwise replacements occur in the two environments.

### 6.4 Limitations

The experiments compare complete algorithms rather than individual design components. In particular, some bandit algorithms differ at the same time in their reduction, base learner, estimator, exploration mechanism, and update rule. Their finite-horizon differences therefore cannot be attributed confidently to a single cause.

More controlled experiments would be needed to isolate such effects. For example, one could hold the base learner fixed while changing the reduction, or vary the reward process while measuring how often profitable pairwise replacements occur.

The empirical comparison is also limited to two one-player environments and two closely related repeated games, with eight replicates and horizon  $T = 10^6$ . The observed trajectories therefore support finite-horizon comparisons, but they do not establish universal rankings or asymptotic behavior beyond the corresponding theoretical guarantees.

## 7 Conclusion

This thesis compared regret-minimizing algorithms under full-information and bandit feedback, with particular attention to finite-horizon swap regret and equilibrium convergence in repeated games.

The experiments show that the distinction between external and stronger regret notions is not only theoretical. Algorithms designed to control action-dependent deviations can behave substantially differently from standalone external-regret learners at finite horizons. At the same time, the strength of this difference depends on the reward sequence, and algorithms with similar theoretical objectives need not have the same finite-horizon performance. The experiments therefore support the theoretical classification of the algorithms without suggesting a universal empirical ranking.

The relative finite-horizon behavior also differs between the full-information and bandit experiments, although the present design does not isolate the effect of feedback from the accompanying algorithmic changes. Across both settings, the tested environment affects how clearly the distinction between regret classes becomes visible.

In self-play, the regret comparison is reflected in the equilibrium behavior of the empirical joint distribution. External-regret control is associated with the weaker CCE requirement, while stronger action-dependent regret control supports convergence toward CE. The experiments therefore connect the algorithmic comparison at the regret level with its game-theoretic consequence.

Taken together, the results show that theoretical regret guarantees capture important qualitative differences between the algorithms, but do not determine their finite-horizon ranking. The practical comparison therefore remains algorithm- and setting-dependent.

### 7.1 Future Work

A first extension would be to study the effect of the feedback model more directly. The present experiments compare full-information and bandit algorithms, but several algorithmic components change together with the feedback model. More controlled experiments on the same reward sequences could separate the loss caused by partial feedback from differences in algorithm design.

A second direction is to study empirical regret scaling more systematically. Running the algorithms over different horizons and, in particular, larger action spaces would make it possible to compare their practical dependence on  $T$  and  $K$  with the corresponding theoretical bounds. Larger symmetric zero-sum games would also provide a stronger test of the difference between Exp3-type external-regret algorithms and swap-regret methods such as

BM and Ito under bandit feedback.

The relation between regret and equilibrium convergence could also be studied in more detail. The current self-play experiments show the expected qualitative difference between CCE and CE convergence, but a broader set of games could test how closely the speed of regret reduction is related to the distance of empirical play from the corresponding equilibrium set.

Finally, the algorithm comparison could be extended to more recent no-swap-regret methods. Evaluating newer algorithms within the same framework would allow their finite-horizon behavior to be compared directly with the reductions and specialized procedures studied in this thesis. Additional diagnostics, such as action frequencies or pairwise replacement opportunities, could then be used when needed to explain observed differences.

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